PHY 103 COMPILED E TEST QUESTIONS

1, The force of attraction between the molecules of different substance is called A, intermolecular force B. cohesive force C, suface tension D. adhesive force

2, ……is the ability of an object to float when it is placed in fluid A. archimede's principle B. relative density C. density D. bouyancy

3.The following are evidence of surface tension except A. a greased iron needle placed gently on the surface of water float on it B. When mercury is split on a clean glass plate it forms globules C. Rain drops are spherical in shaped D. oil drop spreads on hot water, whereas it may remain as a drop on cold water

4. An object weighs 90N and 86N in water find the buoyant force of the water

A. 4N B.1.05N C.0.4N D.0.96N

5. A large metal block displaces 1m3 of water find the buoyant force of the water exerted on the block

A.9800N B.4900N C.102N D.0.0098N

6. Calculate the force required to pull away a horizontal circular loop of wire of radius 0.02m from the surface of water the surface tension of the water is 0.075N/M A.18.8N B.0.188N C.1.88N D.0.0188N

7. A block of metal with a volume of 0.1m3 has a mass of approximately 300kg what is the density of the metal A.3kg/m2 B.300kg/m3 C.30kg/m3 D.3000kg/m3

8. A circular loop of wire of radius 0.004m is lowered into a portion of crude oil. The force due to the surface tension of the oil is 0.01N calculate the surface tension of crude oil A.0.002N/m B.2.00N/m C.0.02N/m D.0.2N/m

9.if the radius R1 of a soap bubble is six times that of another soap bubble of radius R2 what is the excess pressure A.2:6 B.6:1 C.1:6 D.6:2

`10.The force of attraction or repulsion acting between two molecules is known as A.cohesive force B.adhesive force C. intermolecular force D. surface tension

11. The principle state that when a body is totally or partially immersed in fluid it experiences an upthrust in weight to fluid displaced A. Archimede's B. bouyancy C. floatation D. upthrust

12. When a tube of very narrow bore is dipped in a liquid the liquid either ascend or descends the tube this phenonomenon is called

A. surface tension B. angle of contact C. capillarity D. excess pressure

13. The following are example of angle of contact except A. plane meniscus B. convex meniscus C. concave meniscus D. curved meniscus

14. Fluid resist change of volume but not change of shape the statement is A. partially true B. partially false C. true D. false

15. If a body remains the deforming shape or size when deforming force is removed A. perfectly deformation B. perfectly inelastic body C. perfectly plastic body D. perfect rigid body

16. Which of the following term is appropriate for fluids A. shear elasticity B. bulk elasticity C. shape elasticity D. volume elasticity

17. which of the following is the unit of strain A. unitless B. Nm C. N/m2 D. Nm2

18. An elastic limit is reached tensile strain A. decrease more rapidly B. increase in proportion to stress C. decrease in proportion to stress D. increase more rapidly

19. which of the following is related to the amount of deformation produced A. modulus B. poison ratio C. strain D. stress

20. substance which break just after the elastic limit is attained is said to be A. rigid B. ductile C. plastic D. brittle

21. The ratio of normal strain to volume strain is A. rigidity modulus B. bulk modulus C. young modulus D. shear modulus

22. when too many people stand on a bridges it collapses why A. due to friction B. due to overweight C. due to increase in stress D. due to improper construction

23. Which of the following substance has the highest elasticity A. rubber B. sponge C. steel D. copper

24. A liquid has only …….A. young modulus B. shear modulus C. bulk modulus D. All of the above

25. The molecules of solids are arranged in a repeated regular pattern known as A. crystal B. edge C. lattice D. shape

26. Bulk modulus of elasticity corresponds to A. shearing strain B. volume strain C. longitudinal strain D. tensile strain

27. The constant E in hooke's law that relate stress and strain is called A. stiffness constant B. stress constant C. modulus of elasticity D. strain constant

28. Which of the state of matter possess both rigidity and volume elasticity A. gas B. liquid C. solid D. both solid and liquid

29. The change in volume of gas or liquid to the original volume is A. shearing strain B. rigidity strain C. tensile strain D. bulk strain

30. When a rope is pulled on either side what is the stress acting on it A. tensile stress B. compressive stress C. normal stress D. tagential stress

31. The force of an elastic material id directly propotional to the extension produced provided that the elastic is not exceed this statement is known as A. bulk modulus B. hook's law C young modulus D Shear modulus

32. 0bject which when stretched or compressed by a force tend to change their shape but regain it after withdrawal of the force are called A. elastic B compressible C. extensible D stretchable

33. On a force extension graph the slope represent which of the following A. modulus B compressibility C. spring constant D. energy per unit volume

34. The ratio of the lateral strain to the longitudinal strain is called what A. longitudinal ratio B poison ratio C. lateral ration D modulus ratio

35. The reciprocal of bulk modules of elasticity is called A. Rigidity B. compressibility C. ductility D. brittleness

36. The restoring force per unit area is called A. stress B. strain C young modulus D tensile strain

37. If the surface tension of water is 0.08N/m what is the capillarity rise in a tube of diameter of 1mm assuming the angle of contact is θ A. 0.33m B.0.0033m C.3.30mD. 0.033m

38 th following are true for angle of contact except A. soluble impurities increase angle of contact B. partially soluble impurities decreases the angle of contact C. when temperature increase the angle of contact decreases D. angle of contact is higher between pure water and glass than between water and sliver

39. A wooden stick 2m long is floating on the surface of water, the surface tension of water is 0.07N/m by putting soap solution on one side of the stick the surface tension is reduced to 0.06N/m what is the net force of the stick A.0.2N B.0.002N C.2.00N D.0.02N

40. A rectangular film of liquid is formed in a frame of wire and a movable rod of length 4cm what force must be applied to the rod to keep it in equilibrium if the surface tension of the liquid is 40×10^{-3}N/m A. 3.2×10^{-3}N B. 3.2×10^{-2}N C. 3.2×10^{-1}N D. 3.2×10^{3}N